

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk:

Product Description:

Cat No. :

SurePrep™ RNA/Protein Purification Kit, Protein pH Binding Buffer

SurePrep™ RNA/Protein Purification Kit, Protein pH Binding Buffer

BP2806-50F

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use

Laboratory chemicals.

Uses advised against

No Information available

**Company**

 Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
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 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**E-mail address**

Enquiry.my@thermofisher.com

**Emergency Telephone Number**

Tel: +03-5525 7888

 CHEMTREC Malaysia **1-800-815-308** (Malay)

 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                        |                   |
|----------------------------------------|-------------------|
| Substances/mixtures corrosive to metal | Category 1 (H290) |
| Skin Corrosion/Irritation              | Category 2 (H315) |
| Serious Eye Damage/Eye Irritation      | Category 1 (H318) |

**Label Elements**


Signal Word

Danger

**Hazard Statements**

H290 - May be corrosive to metals

H315 - Causes skin irritation

H318 - Causes serious eye damage

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## Precautionary Statements

### Prevention

P234 - Keep only in original packaging

P262 - Do not get in eyes, on skin, or on clothing

P264 - Wash face, hands and any exposed skin thoroughly after handling

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P331 - Do NOT induce vomiting

P390 - Absorb spillage to prevent material damage

P362 + P364 - Take off contaminated clothing and wash it before reuse

### Storage

P402 - Store in a dry place

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P406 - Store in corrosion resistant polypropylene container with a resistant inliner

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component            | CAS No    | Weight % |
|----------------------|-----------|----------|
| Nuclease Free Water  | 7732-18-5 | 70 - 80  |
| Citric acid          | 77-92-9   | 15 - 20  |
| Orthophosphoric acid | 7664-38-2 | 8 - 10   |
| Sodium hydroxide     | 1310-73-2 | < 5      |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

#### Skin Contact

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes.

#### Ingestion

Clean mouth with water and drink afterwards plenty of water. If symptoms persist, call a physician.

#### Inhalation

Remove to fresh air.

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Most important symptoms and effects, both acute and delayed

Causes severe eye damage. .

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## Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Substance is nonflammable; use agent most appropriate to extinguish surrounding fire.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

### **Hazardous Combustion Products**

Oxides of phosphorus.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing.

### Environmental precautions

Do not flush into surface water or sanitary sewer system.

### Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place.

### Specific End Uses

Use in laboratories.

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## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component            | Malaysia | ACGIH TLV                                             | OSHA PEL                                                                                              |
|----------------------|----------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Orthophosphoric acid |          | TWA: 1 mg/m <sup>3</sup><br>STEL: 3 mg/m <sup>3</sup> | (Vacated) TWA: 1 mg/m <sup>3</sup><br>(Vacated) STEL: 3 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> |
| Sodium hydroxide     |          | Ceiling: 2 mg/m <sup>3</sup>                          | (Vacated) Ceiling: 2 mg/m <sup>3</sup><br>TWA: 2 mg/m <sup>3</sup>                                    |

| Component            | European Union                                                     | The United Kingdom                                    | Germany                                                                                                                                      |
|----------------------|--------------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Citric acid          |                                                                    |                                                       | TWA: 2 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 2 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 4 mg/m <sup>3</sup> |
| Orthophosphoric acid | TWA: 1 mg/m <sup>3</sup> (8h)<br>STEL: 2 mg/m <sup>3</sup> (15min) | STEL: 2 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | TWA: 2 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 2 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 4 mg/m <sup>3</sup> |
| Sodium hydroxide     |                                                                    | 2 mg/m <sup>3</sup> STEL                              | 2 mg/m <sup>3</sup> TWA (inhalable fraction)                                                                                                 |

### Exposure Controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Protective gloves

#### Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

#### Recommended Filter type:

Particulates filter conforming to EN 143

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

### Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water system

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

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## Information on basic physical and chemical properties

|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Appearance                              | Clear Colorless          |                                   |
| Physical State                          | Liquid                   |                                   |
| Odor                                    | No information available |                                   |
| Odor Threshold                          | No data available        |                                   |
| pH                                      | 1.5                      |                                   |
| Melting Point/Range                     | No data available        |                                   |
| Softening Point                         | No data available        |                                   |
| Boiling Point/Range                     | No information available |                                   |
| Flash Point                             | Not applicable           | Method - No information available |
| Evaporation Rate                        | No data available        |                                   |
| Flammability (solid,gas)                | Not applicable           | Liquid                            |
| Explosion Limits                        | No data available        |                                   |
| Vapor Pressure                          | No data available        |                                   |
| Vapor Density                           | No data available        | (Air = 1.0)                       |
| Specific Gravity / Density              | No data available        |                                   |
| Bulk Density                            | Not applicable           | Liquid                            |
| Water Solubility                        | No information available |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Component                               | log Pow                  |                                   |
| Citric acid                             | -1.72                    |                                   |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | No data available        |                                   |
| Viscosity                               | No data available        |                                   |
| Explosive Properties                    | No information available |                                   |
| Oxidizing Properties                    | No information available |                                   |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

|                          |                           |
|--------------------------|---------------------------|
| Hazardous Polymerization | No information available. |
| Hazardous Reactions      | No information available. |

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## Conditions to Avoid

Incompatible products. Excess heat.

## Incompatible Materials

Strong oxidizing agents.

## Hazardous Decomposition Products

Oxides of phosphorus.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

**Product Information** No acute toxicity information is available for this product

**(a) acute toxicity;**

**Oral**

Based on available data, the classification criteria are not met

**Dermal**

Based on available data, the classification criteria are not met

**Inhalation**

Based on available data, the classification criteria are not met

### Toxicology data for the components

| Component            | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation                   |
|----------------------|---------------------------|------------------------------|-----------------------------------|
| Nuclease Free Water  | LD50 > 90 mL/kg ( Rat )   | -                            | -                                 |
| Citric acid          | LD50 = 3 g/kg ( Rat )     | >2 g/kg ( Rat )              | -                                 |
| Orthophosphoric acid | LD50 = 1530 mg/kg ( Rat ) | LD50 = 2740 mg/kg ( Rabbit ) | 850 mg/m <sup>3</sup> ( Rat ) 1 h |
| Sodium hydroxide     | LD50 = 325 mg/kg ( Rat )  | LD50 = 1350 mg/kg ( Rabbit ) | -                                 |

**(b) skin corrosion/irritation;** Category 2

**(c) serious eye damage/irritation;** Category 1

**(d) respiratory or skin sensitization;**

**Respiratory**

No data available

**Skin**

No data available

**(e) germ cell mutagenicity;** No data available

**(f) carcinogenicity;** No data available

There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;** No data available

**(h) STOT-single exposure;** No data available

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(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Other Adverse Effects The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed No information available.

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects** Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component            | Freshwater Fish                                     | Water Flea             | Freshwater Algae       | Microtox                                          |
|----------------------|-----------------------------------------------------|------------------------|------------------------|---------------------------------------------------|
| Citric acid          | Leuciscus idus: LC50 = 440-760 mg/L/96h             | EC50 = 120 mg/L/72h    |                        | Photobacterium phosphoreum: EC50 = 14 mg/L/15 min |
| Orthophosphoric acid | 98 - 106 mg/L LC50 96 h                             | > 100 mg/L EC50 = 48 h | > 100 mg/L EC50 = 72 h |                                                   |
| Sodium hydroxide     | LC50: = 45.4 mg/L, 96h static (Oncorhynchus mykiss) | -                      | -                      | -                                                 |

**Persistence and degradability** No information available  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative potential** No information available

| Component   | log Pow | Bioconcentration factor (BCF) |
|-------------|---------|-------------------------------|
| Citric acid | -1.72   | No data available             |

**Mobility in soil** .

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

**Waste treatment methods**  
**Waste from Residues/Unused Products** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification Waste

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is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

## Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

## Other Information

Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Solutions with low pH-value must be neutralized before discharge

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN1805  
Hazard Class 8  
Packing Group III  
Proper Shipping Name PHOSPHORIC ACID SOLUTION

### Road and Rail Transport

UN-No UN1805  
Hazard Class 8  
Packing Group III  
Proper Shipping Name PHOSPHORIC ACID, SOLUTION

### IATA

UN-No UN1805  
Hazard Class 8  
Packing Group III  
Proper Shipping Name PHOSPHORIC ACID, SOLUTION

### Special Precautions for User

No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

X = listed

| Component            | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|----------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Nuclease Free Water  | 231-791-2 | X    | X   | X     | X    |      | X     | X    | KE-35400 |
| Citric acid          | 201-069-1 | X    | X   | X     | X    | X    | X     | X    | KE-20831 |
| Orthophosphoric acid | 231-633-2 | X    | X   | X     | X    | X    | X     | X    | KE-27427 |
| Sodium hydroxide     | 215-185-5 | X    | X   | X     | X    | X    | X     | X    | KE-31487 |

| Component            | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Citric acid          |                                                                                           |                                                                                          |                            | Annex I - Y34                      |
| Orthophosphoric acid |                                                                                           |                                                                                          |                            | Annex I - Y34                      |
| Sodium hydroxide     |                                                                                           |                                                                                          |                            | Annex I - Y35                      |

### National Regulations



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**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Revision Date**

24-Mar-2025

**Revision Summary**

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**